THAKUR VAIBHAV SINGH

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Portfolio: https://vaibhavthkr.github.io/portfolio/

EDUCATION

University College of Engineering (A), Osmania University — Hyderabad, TG
B.E Biomedical Engineering — August 2023 – June 2027
Relevant Coursework: Biomedical Equipment; Biomedical Signal Processing, Biomedical Image processing, Biomedical Imaging systems, Biomaterials, Biomedical Equipment; Medical Device Regulations

Sri Chaitanya Junior College — ECIL, Hyderabad
Mathematics, Physics, Chemistry — April 2020 – May 2023

Nalanda High School — Neredmet, Hyderabad
SSC — April 2020

PROJECTS

A Deep Learning based Real-Time Estimation of Tibiofemoral Contact Forces using Wearable IMU Sensors and Personalized Neuromusculoskeletal Modelling

- Currently working on it as Academic mini project
- Estimate real time knee joint contact forces based upon the knee joint angle using IMU sensors and NMSM modelling.

Fabrication of PCL nanofiber using air blown method

- Fabricated PCL Nanofibers
- Used Air compressor to generate high pressure to spray PCL Solution

Low cost portable tribometer

- Made a simple pin on disk tribometer.
- Integrated speed control and timer for the stepper motor, the deflections from the stylus are recorded through the load cell and are plotted in the excel.

Computer vision based upper limb rehabilitation system

- Made a computer vision-based GUI using python for rehabilitation of wrist joint, elbow joint and shoulder joint.
- It consists of demo video of exercise and real time visual and audio feedback, reps counter, exercise history.

Real-time gait parameters monitoring via MPU6050 and ESP32

- Monitored various parameters such as Heel strike, stance period, toe off etc.
- The system was wireless just attach on the Heel and use, data was plotted in MATLAB

Gamified computer vision-based palm rehabilitation system

- The patients can play Dino game using their palm
- This project helps gamification of Rehabilitation systems; further research could involve designing personalized gamified Rehabilitation systems

SKILLS

Software & Simulation: MATLAB, Python, C, SolidWorks (CAD), Proteus, ImageJ, Arduino IDE, Keil μvision, OpenSim, Canva, Mimics.
Electronics & Embedded Systems: ESP32, Arduino Uno, IoT (Internet of Things), Embedded Systems, Biomedical Sensors, Soldering, Circuit Design, Wearable devices.
Biomaterials & Lab Skills: Nanofiber Fabrication, Biotribology, Biomaterials Characterization, 3D Printing (Biomedical Prototyping & Anatomical Models).
Core Competencies: Biomedical Signal Processing, Biomedical Equipment Management, Medical Image processing.
Soft Skills: Communication, Problem solving, Presentation, Team work, Analytic, Documentation.

ACTIVITIES

IEEE SB OUCE EMBS
Organizer — Apr 2025 – (Present) Apr 2026

- Organized and coordinated for 10+ events

IEEE SB OUCE
Design Lead — Apr 2025 - (Present) Apr 2026

- Managing and guiding design team of 12 design volunteers for graphic design.

National Symposium on Instrumentation - 45, ISOI Bangalore
Volunteer — Dec 2025 - Jan 2026

- Lead the Design team, delivered Brochure, invitations, various posters, certificates, ID Cards.
- Stage management and ensured smooth flow of the event

TEDxOsmaniaU 2025
Committee Member — Aug 2025 – Nov 2025

- Received the Letter of Recommendation from the Managing Director, TEDxOsmaniaU 2025
- Handled Stage management during the main event and also the Theme reveal event of the TEDxOsmaniaU 2025.

Meditech 2k25
Design and Operations Organizer — Jan 2025 - Mar 2025

- Was a Design Volunteer and Operations member

Section Student Congress 2025, IEEE Hyderabad Section
Volunteer — Oct 2024

- Volunteered in logistics of the event

Meditech 2k24
Design Volunteer — Jan 2024 - Mar 2024

- Volunteered for design for various posters, invitations, Brochure, Certificates

ACHIEVEMENTS/ CO CURRICULAR ACTIVITIES

- Achieved GATE 2026 **AIR 154** in Biomedical Engineering paper. Score card
- Volunteered in organizing National Level Hands on Workshop on Building Smart Biomedical Devices using IOT on 20th – 21st February 2026.
- Won 1st prize in Poster presentation competition with my teammate Ushasree B, presented "Mitigating SANS A Novel Wearable Device for Real-Time Non-Invasive ICP Monitoring in Spaceflight" in National Symposium on Instrumentation-45 organized by Department of Biomedical Engineering, UCE (A), OU in collaboration with ISOI, IISc Bangalore. Certificate link
- Presented a paper "A Multimodal Control Approach for Smart Wheelchairs Using Accelerometer and EMG Gesture Interfaces" in IFIP IOT'25. Certificate link
- Attended Section Student Congress organized by IEEE Hyderabad section. Certificate link
- Attended Hands on training on Biomedical Equipment organized by department of Biomedical Engineering, UCE (A), OU.
- Participated in National Level Technical Symposium Promethean 2025 organized by BV Raju Institute of Technology, Narsapur, Telangana.
- Won 1st Prize in Poster presentation competition, presented "Thermal vision-based Sleep apnea detection using CNN and LSTM networks" with my teammates Hema Harshitha and Beulah Sadhvi E. in Nation level Technical Symposium organized by Spoorthy Engineering college, Hyderabad. Certificate link
- Attended Hands on workshop on Biomaterials organized by department of Biomedical Engineering, UCE (A), OU.
- Participated in the National Seminar on Recent Trends in Medical Equipment – 24 (RTME-24) conducted by Department of Biomedical Engineering, University College of Engineering (A), Osmania University. Certificate link
- Won consolation in Model presentation, presented "Gamified computer vision-based palm rehabilitation system" in Meditech 2K24, organized by Department of Biomedical Engineering, UCE (A), OU. Certificate link
- Won 1st prize in Quiz competition in Hands on Workshop on Biomaterials and Fabrication Techniques organized by Department of Biomedical Engineering, UCE (A), OU. Certificate link